WestQuant Memory Discovery Lab

Executed system evaluation addendum

David Vesterlund | WestQuant | 10 October 2026

Repository: https://github.com/VesterlundCoder/memory-discover-lab

Outcome. The lab agrees with an independently written NumPy/SciPy reference on the declared matched workload. Seven constructed contract faults are blocked with passing positive controls. Local performance and checkpoint tests pass. A second-family control is executable, but a second experimental physical baseline remains pending.

1. Executed matched comparison

Both arms receive identical reviewed conditions. The reference imports no lab channel constructor: it exponentiates the column-vectorized Lindblad generator, then independently constructs the erasure-inclusive Choi matrix. One experiment-constrained optical point and twelve hypothetical Markov profiles cover 73 channel-age points. The two source parsers consume the same SHA256-verified experimental files.

Metric	Executed result	Gate
Profiles / channel-age points	13 / 73	Specified workload
Full Choi maximum error	1.3103e-14	<= 1e-8
Six-state output maximum error	1.1828e-14	<= 1e-8
Conditional fidelity maximum error	3.9968e-15	<= 1e-8
Acceptance maximum error	0	<= 1e-8
Processed fidelity scalars	90; maximum discrepancy 0	<= 1e-8

Experimental file	Values	Mean (both arms)
4.csv	30	0.9510824229
8a.csv	60	0.9140479475

The source is Teller et al., Zenodo record 14929997, version v1 (DOI 10.5281/zenodo.14929997). These are released processed fidelities, not independent raw-count measurements. The measured optical operating point does not identify physical spin T2.

Local construction timing

Five randomized-order warm repeats per arm: median lab 0.05438 s and direct reference 0.09963 s. Export, metadata and validation costs are not equalized; file IO, fitting, network events and downstream solvers are excluded. These are descriptive local timings, not evidence of end-to-end speed superiority.

The configuration, seed, actual timing order, input hashes, package environment, thread settings and source fingerprints are retained in evaluation/results/step1-executed. Results are retrospective, not prospectively preregistered.

Interpretation and existing workflows

Agreement verifies that integration preserves the supplied equations and processed data. It does not establish new physics, a new material lifetime, less researcher effort or general superiority over custom scripts. General workflow engines can host the same domain contracts.

Tool	Existing capability / relationship	Step-1 execution
NumPy/SciPy script baseline	Explicit data parsing and supplied Lindblad/channel equations Independent executable baseline; equivalent reviewed input assumptions	Executed matched numerical comparison
QuTiP	Open-system evolution and channel representations Underlying solver; memory evidence and device contracts supplied by integration code	Documentation audit in step 1; earlier article controls are separate
PyCCE	CCE/gCCE spin-bath dynamics and supplied couplings Underlying spin solver; model convergence and material inputs remain application responsibilities	Documentation audit only in step 1
PySCF	Electronic-structure calculations Underlying electronic solver; electronic results do not by themselves identify a lifetime or optical instrument	Documentation/context only in step 1
SeQUeNCe	Event-driven quantum-network components, protocols and applications Downstream network engine; WestQuant exports a declared memory channel	Documentation audit in step 1; earlier article workload is separate
AiiDA	General provenance graphs, workflow orchestration and calculation caching Potential host for WestQuant domain contracts; provenance and caching are not claimed as new	Documentation audit only; AiiDA was not run

AiiDA already supports provenance and caching. It was audited through its documentation, not installed and benchmarked against this lab. The contribution evaluated here is reusable quantum-memory condition, evidence, age and channel semantics. The table is a scoped documentary comparison, not an exhaustive novelty claim.

Primary documentation

NumPy/SciPy script baseline: https://docs.scipy.org/doc/scipy/reference/generated/scipy.linalg.expm.html

QuTiP: https://qutip.readthedocs.io/en/stable/guide/guide.html

PyCCE: https://pycce.readthedocs.io/en/latest/tutorial.html

PySCF: https://doi.org/10.1063/5.0006074

SeQUeNCe: https://github.com/sequence-toolbox/SeQUeNCe

AiiDA: https://aiida.readthedocs.io/projects/aiida-core/en/stable/topics/provenance/index.html

Matched conventions

Qubit states are column-vectorized; frequencies are in Hz and evolution includes 2*pi. Population relaxation and independent pure dephasing are separate rates. Depolarization uses E(rho)=(1-p)rho+p*I/2. State-independent retrieval failure is represented by an orthogonal erasure outcome. Choi matrices are unnormalized with trace equal to input dimension. Absolute-age write-store-read channels are applied once to the original input state.

Additional executed checks and remaining evidence

Step	Implemented and executed	Remaining limitation
2: second family	41-point molecular-spin software control passes (max normalized residual 3.72e-12); hypothetical interface exported with opt-in	Real audited curve, Hamiltonian and optical interface are still required
3: ablations	7/7 invalid contracts blocked; all positive controls pass	Constructed faults, not researcher error incidence
4: performance	Nine cold studies; 108 cold jobs and 108 checkpoint reuses; invariant channel results	Single environment; bounded small models; no claim of large-scale scalability
Tests	75 original + 7 evaluation tests pass	Independent external reproduction pending

Workers	Jobs per study	Cold median [min,max], s	Resume median, s
1	12	3.867 [3.809, 3.910]	0.0181
2	12	1.978 [1.917, 2.067]	0.0174
4	12	1.178 [1.130, 1.229]	0.0182

Each worker uses one native numerical thread. Times include subprocess startup, calculation and artifact IO. Resume reuses terminal-result checkpoints. Parent peak RSS and largest terminated child peak RSS are recorded separately, not as aggregate concurrent memory. Missing-data and actual process-timeout probes pass. Resume from a completed partial checkpoint succeeds; no forced crash or power-loss test was performed.

Reproduce from a fresh clone

git clone https://github.com/VesterlundCoder/memory-discover-lab.git

cd memory-discover-lab

python -m venv .venv

source .venv/bin/activate

python -m pip install -e ./lab

python scripts/evaluate_system.py step1 --out evaluation/results/new-step1

python -m unittest discover -s evaluation/tests -v

First-run step 1 automatically fetches two pinned Zenodo files and verifies their hashes. Full original article reproduction needs optional dependencies and source retrieval; see evaluation/README.md. Unverified third-party caches are absent from the public repository.

Next physical and publication gates

A measured second-family case must provide reviewed complex pure-dephasing observations, uncertainties, model parameters, isotope/composition, tensor frames, temperature, field and pulse conditions. A coherence curve alone cannot validate an optical quantum memory. Independent clean-environment reproduction and scientific review are reserved for the author and colleagues. No hardware experiment, new-material T2 prediction, statistical prediction error margin, better-than-SOTA result, DOI or peer-review acceptance is claimed.